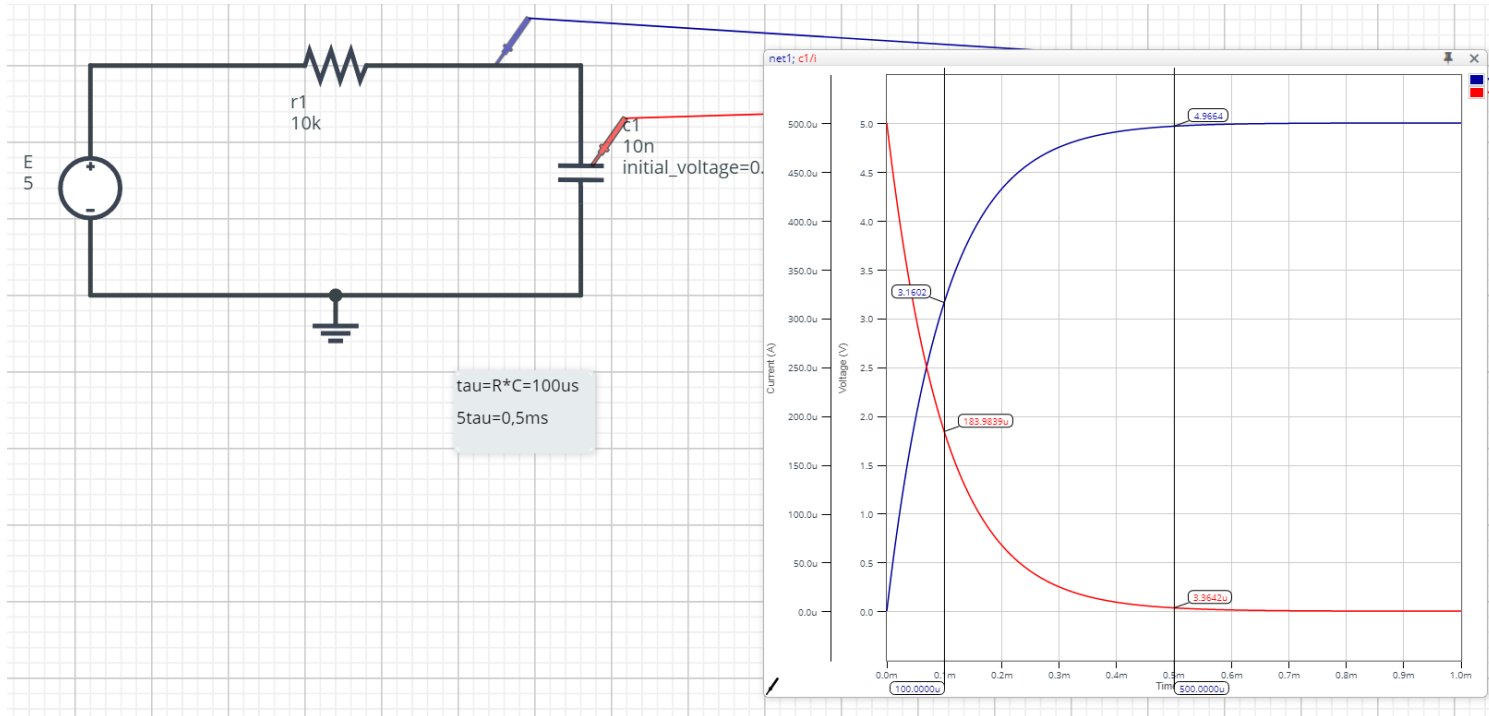


Bazele Circuitelor Electronice

Circuite RC

Comportarea în timp a circuitelor RC

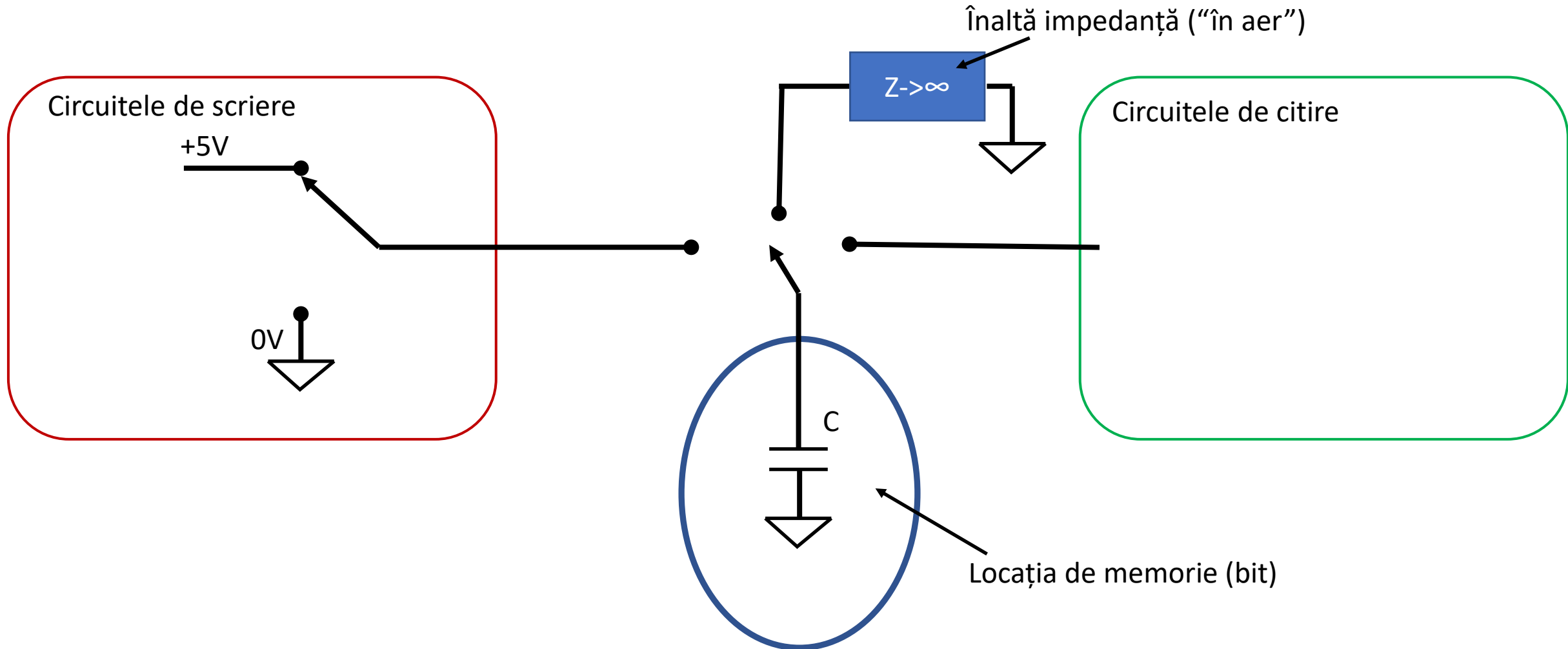


Constanta de timp: $\tau = R \cdot C[s]$

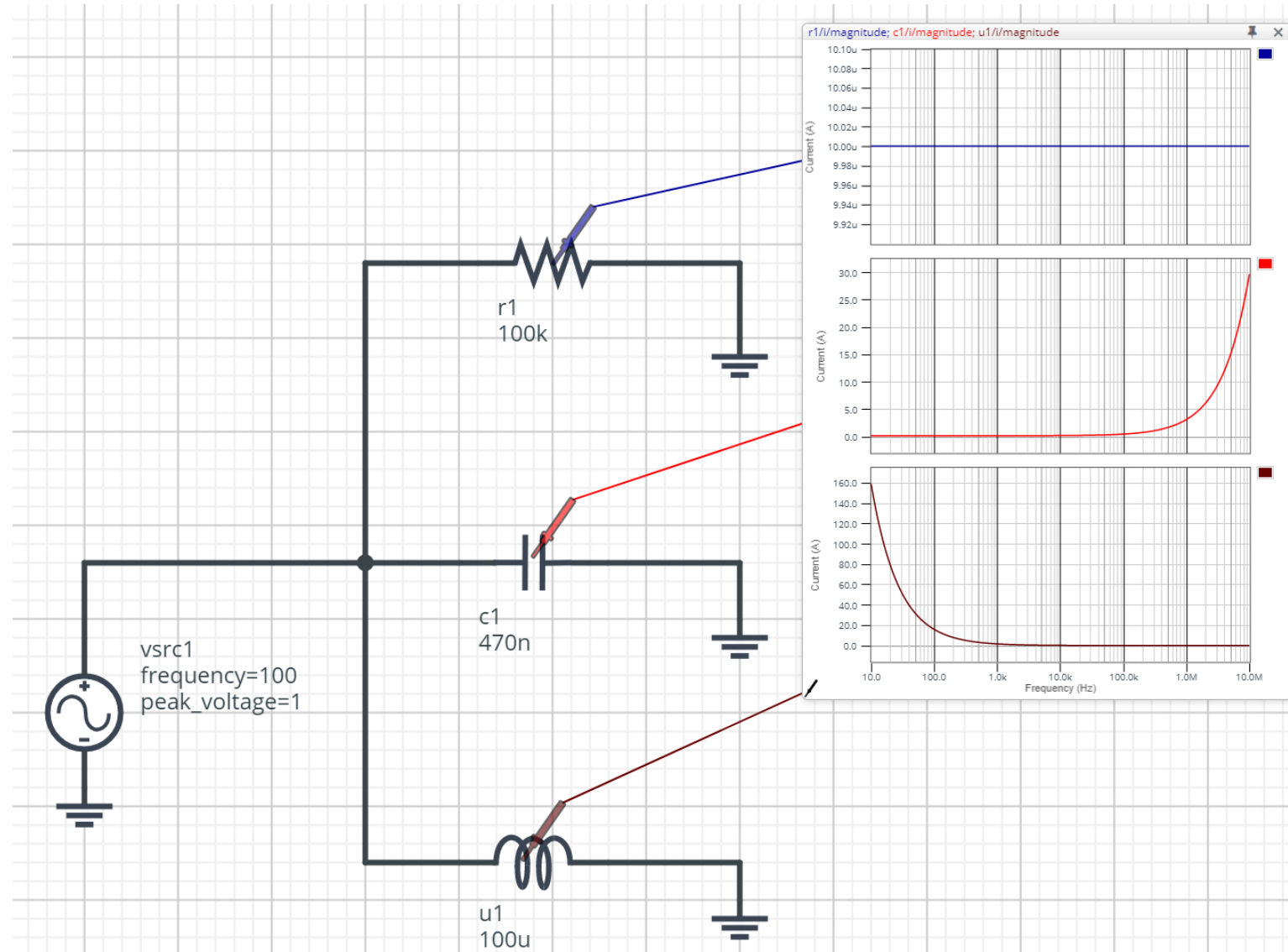
- tensiunea crește de e ori

$$v_c(t) = V_c(\infty) - [V_c(\infty) - V_c(0)] e^{-\frac{t}{\tau}}$$

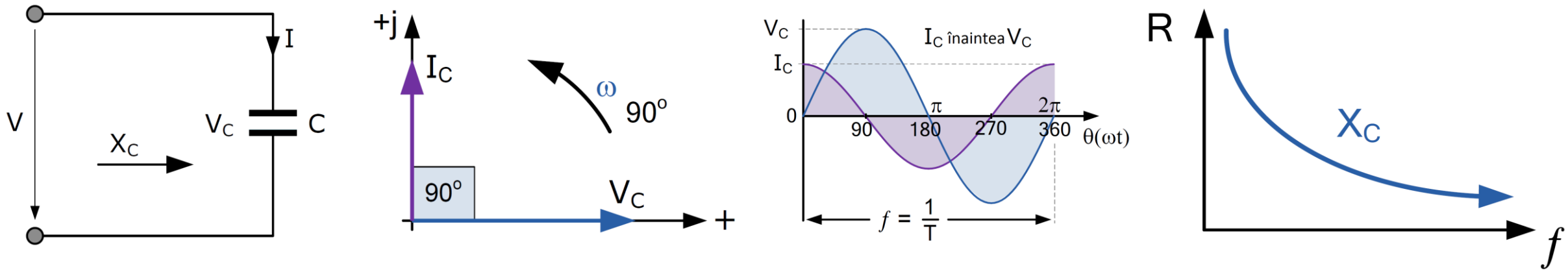
Comportarea în timp a circuitelor RC



Comportarea în frecvență a circuitelor RC

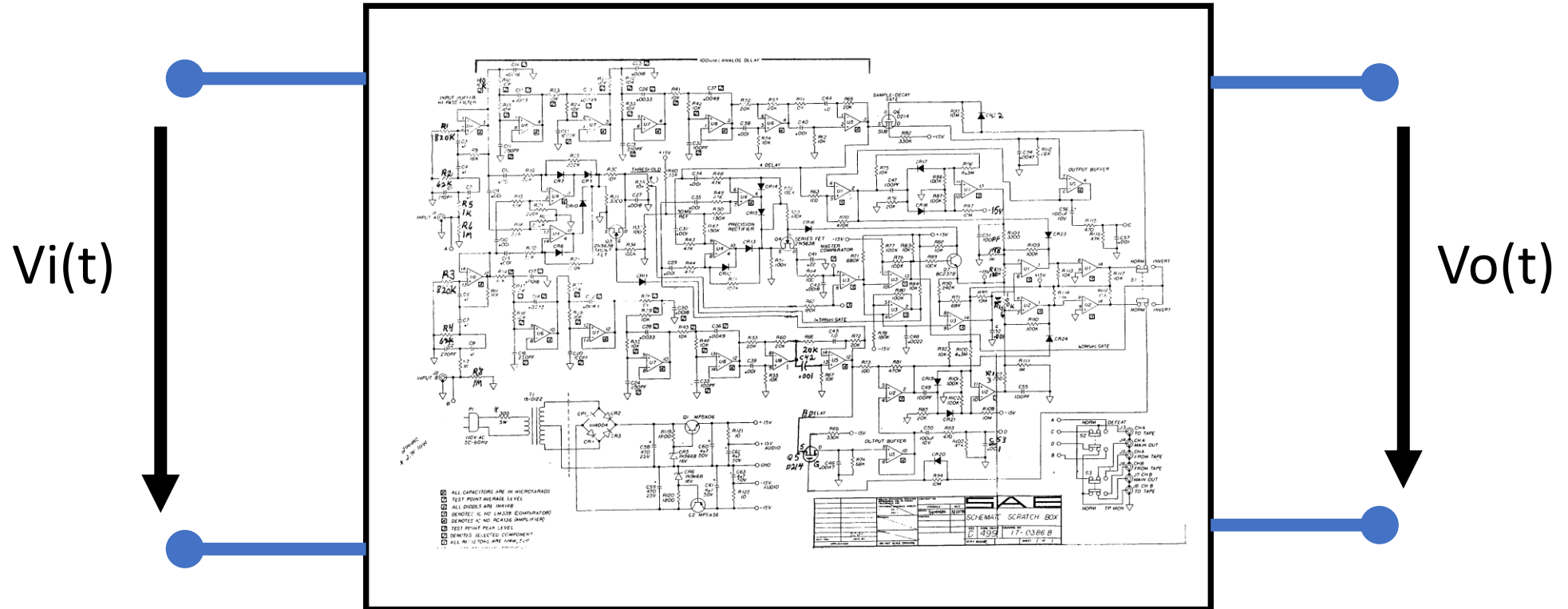


Comportarea în frecvență a circuitelor RC



$X_C = \frac{1}{2\pi f C} = \frac{1}{j\omega C}$		
$f \rightarrow 0 \text{ Hz}$	$X_C \rightarrow \infty \Omega$	$I_C \rightarrow 0 \text{ A}$
$f \rightarrow \infty \text{ Hz}$	$X_C \rightarrow 0 \Omega$	$I_C \rightarrow \infty \text{ A}$
$Z = 0 + j\omega C$		

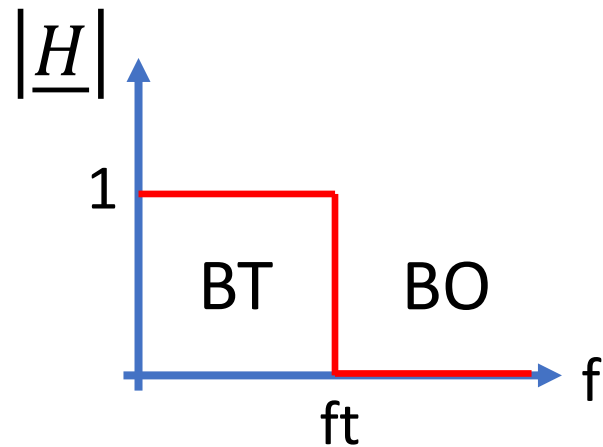
Tipuri de filtre RC



$$\underline{H}(\mathrm{j}\omega) = \frac{\underline{V}_o}{\underline{V}_i}$$

Tipuri de filtre RC

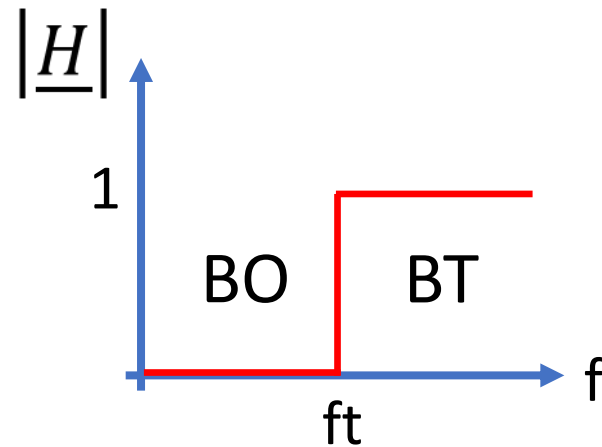
Filtru Trece Jos



$$\varphi = \begin{cases} 0 \text{ grade, } BT \\ \text{inexistent, } BO \end{cases}$$

Tipuri de filtre RC

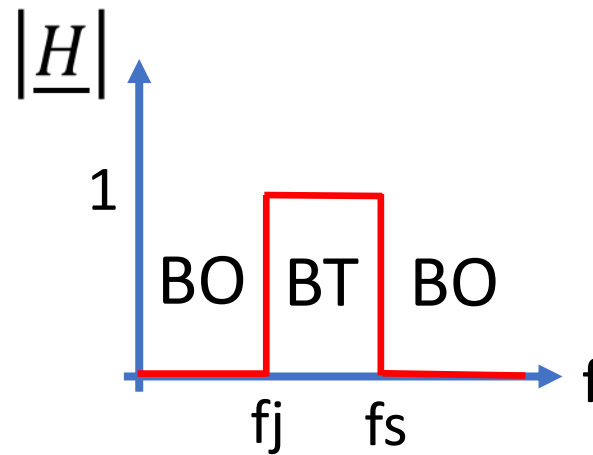
Filtru Trece Sus



$$\varphi = \begin{cases} 0 \text{ grade, } BT \\ \text{inexistent, } BO \end{cases}$$

Tipuri de filtre RC

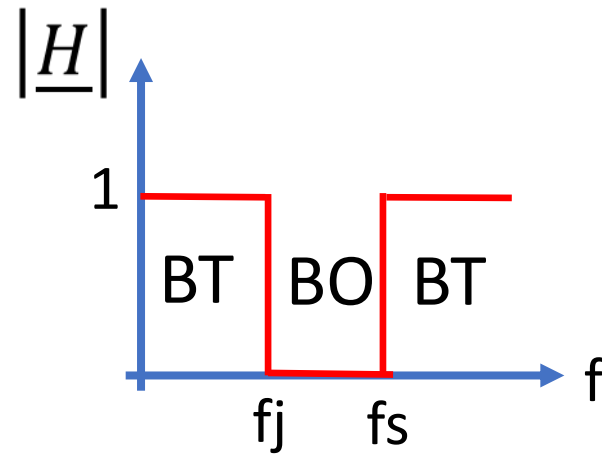
Filtru Trece Bandă



$$\varphi = \begin{cases} 0 \text{ grade}, & BT \\ \text{inexistent}, & BO \end{cases}$$

Tipuri de filtre RC

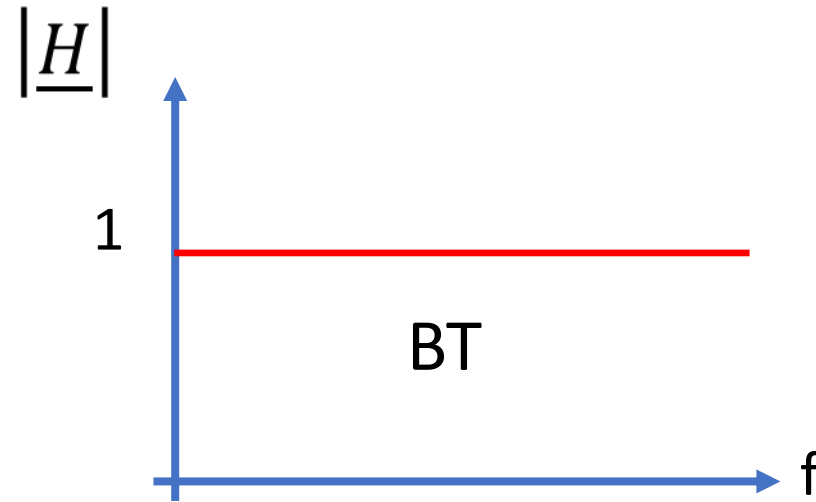
Filtru Oprește Bandă



$$\varphi = \begin{cases} 0 \text{ grade, } BT \\ \text{inexistent, } BO \end{cases}$$

Tipuri de filtre RC

Filtru Trece Tot

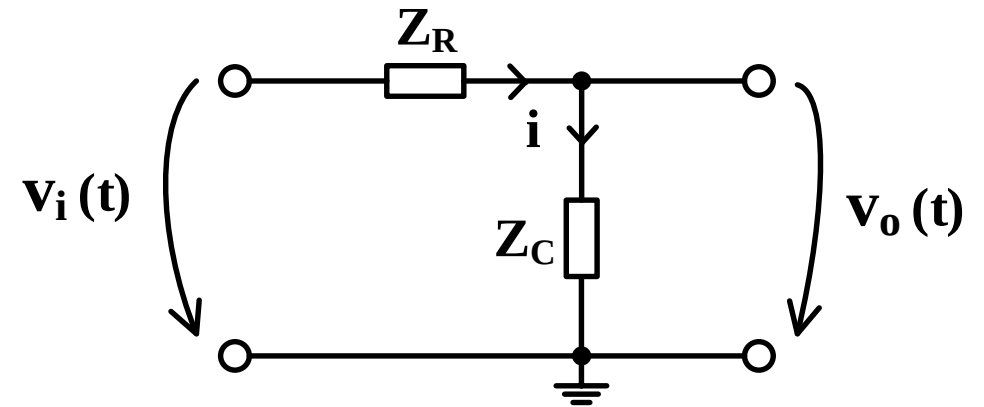
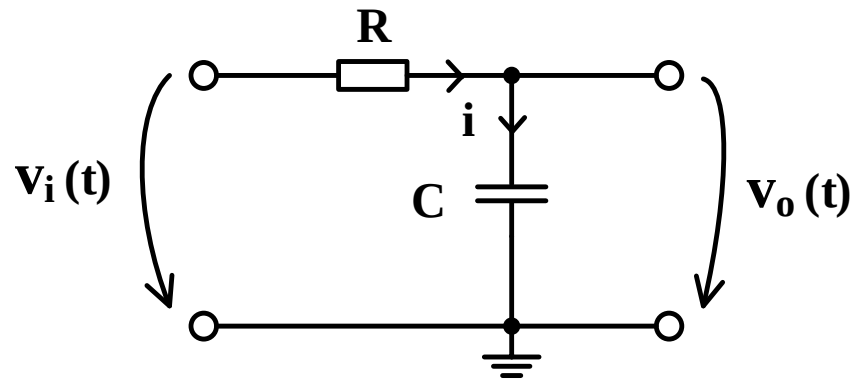


$$\varphi = \begin{cases} 0 \text{ grade, } BT \\ \text{inexistent, } BO \end{cases}$$



Filtrul Trece Jos

I. Trecerea la impedanțe echivalente



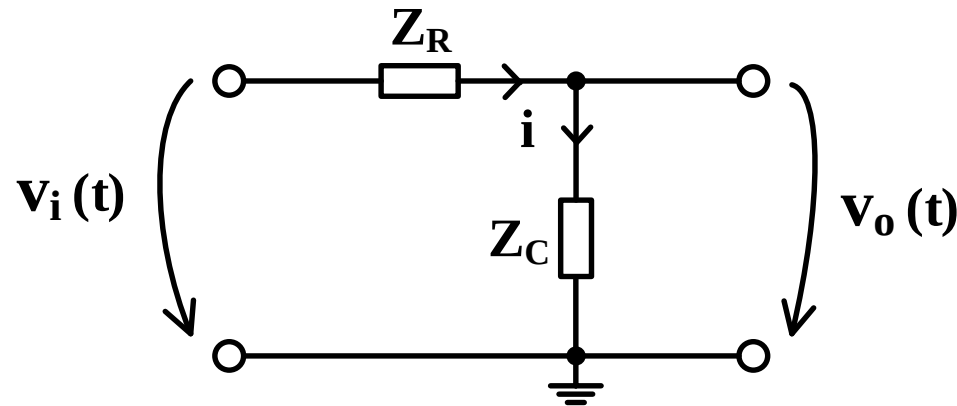
$$Z_R = R$$

$$Z_C = \frac{1}{j\omega C}$$

Filtrul Trece Jos

II. Determinarea funcției de transfer

$$\underline{H}(j\omega) = \frac{\underline{v}_o}{\underline{v}_i}$$



$$\underline{v}_o = \frac{\underline{Z}_C}{\underline{Z}_R + \underline{Z}_C} \cdot \underline{v}_i \Rightarrow \underline{H} = \frac{\underline{v}_o}{\underline{v}_i} = \frac{\underline{Z}_C}{\underline{Z}_R + \underline{Z}_C}$$

Filtrul Trece Jos

III. Înlocuirea impedanțelor

$$\underline{H} = \frac{\underline{Z}_2}{\underline{Z}_1 + \underline{Z}_2} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{\frac{1}{j\omega C}}{\frac{1 + j\omega C R}{j\omega C}} = \frac{1}{1 + j\omega RC}$$

Filtrul Trece Jos

IV. Normarea funcției de transfer

$$f_t = \frac{1}{2\pi RC}$$

$$x = \frac{f}{f_t}$$

$$\underline{H} = \frac{1}{1 + j\omega RC} = \frac{1}{1 + jx}$$

Filtrul Trece Jos

V. Determinarea modului și a fazei funcției de transfer normalată

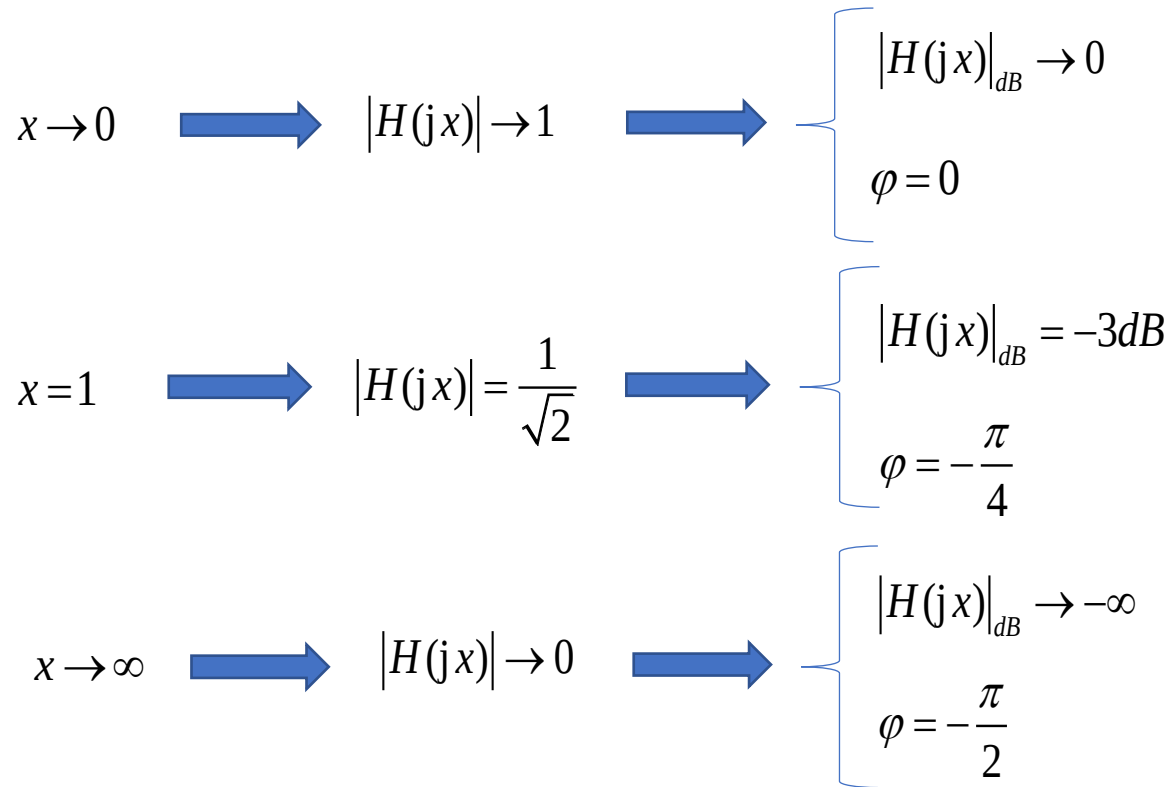
$$\left| \underline{H(jx)} \right| = \left| \frac{1}{1 + jx} \right| = \left| \frac{1 - jx}{(1 - jx) \cdot (1 + jx)} \right| = \left| \frac{1 - jx}{1 + x^2} \right| \Rightarrow \begin{cases} \left| \underline{H(jx)} \right| = \frac{\sqrt{1 + x^2}}{1 + x^2} = \frac{1}{\sqrt{1 + x^2}} \\ \varphi = \arctg(-x) \end{cases}$$

Filtrul Trece Jos

VI. Reprezentarea grafică a modulului și a fazei

$$|\underline{H}|_{dB} = 20 \lg |\underline{H}|$$

$$\Rightarrow \begin{cases} |H(jx)| = \frac{\sqrt{1+x^2}}{1+x^2} = \frac{1}{\sqrt{1+x^2}} \\ \varphi = \arctg(-x) \end{cases}$$



Filtrul Trece Jos

VI. Reprezentarea grafică a modulului și a fazei

$$|\underline{H}|_{dB} = 20 \lg |\underline{H}|$$

$$\Rightarrow \begin{cases} |H(jx)| = \frac{\sqrt{1+x^2}}{1+x^2} = \frac{1}{\sqrt{1+x^2}} \\ \varphi = \arctg(-x) \end{cases}$$

$$x \rightarrow 0 \quad \longrightarrow \quad |H(jx)| \rightarrow 1$$

$$|H(jx)|_{dB} \rightarrow 0$$

$$\varphi = 0$$

$$x = 1 \quad \longrightarrow \quad |H(jx)| = \frac{1}{\sqrt{2}}$$

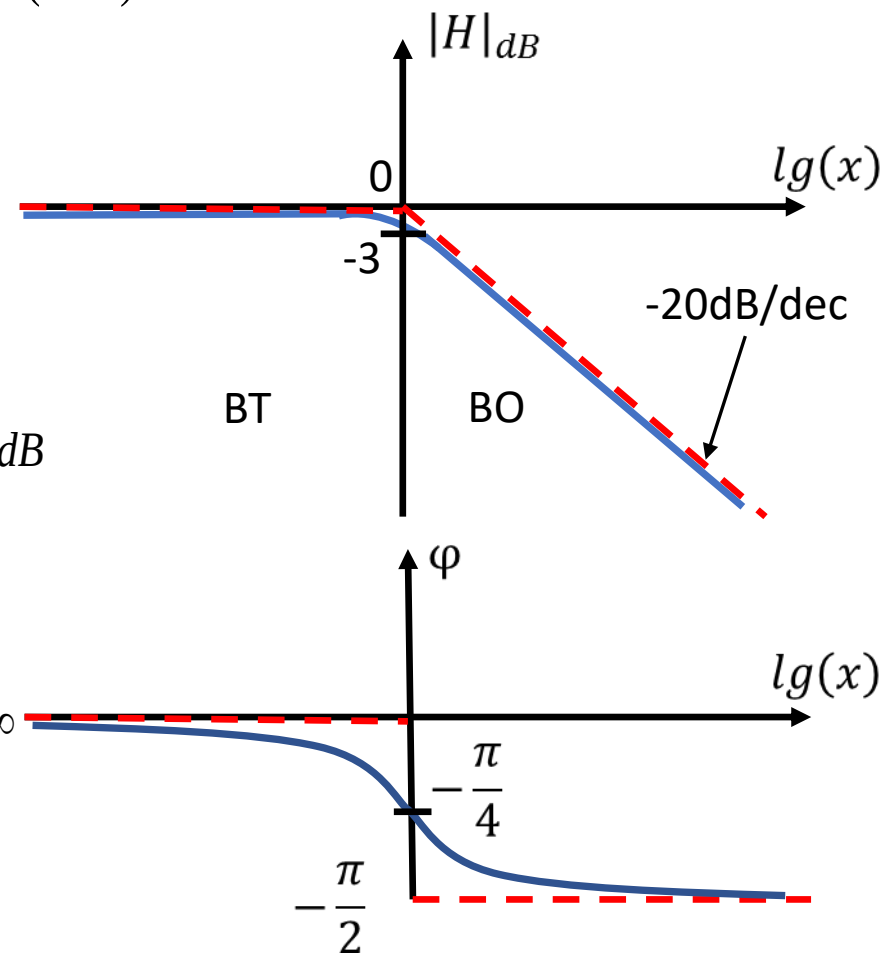
$$|H(jx)|_{dB} = -3dB$$

$$\varphi = -\frac{\pi}{4}$$

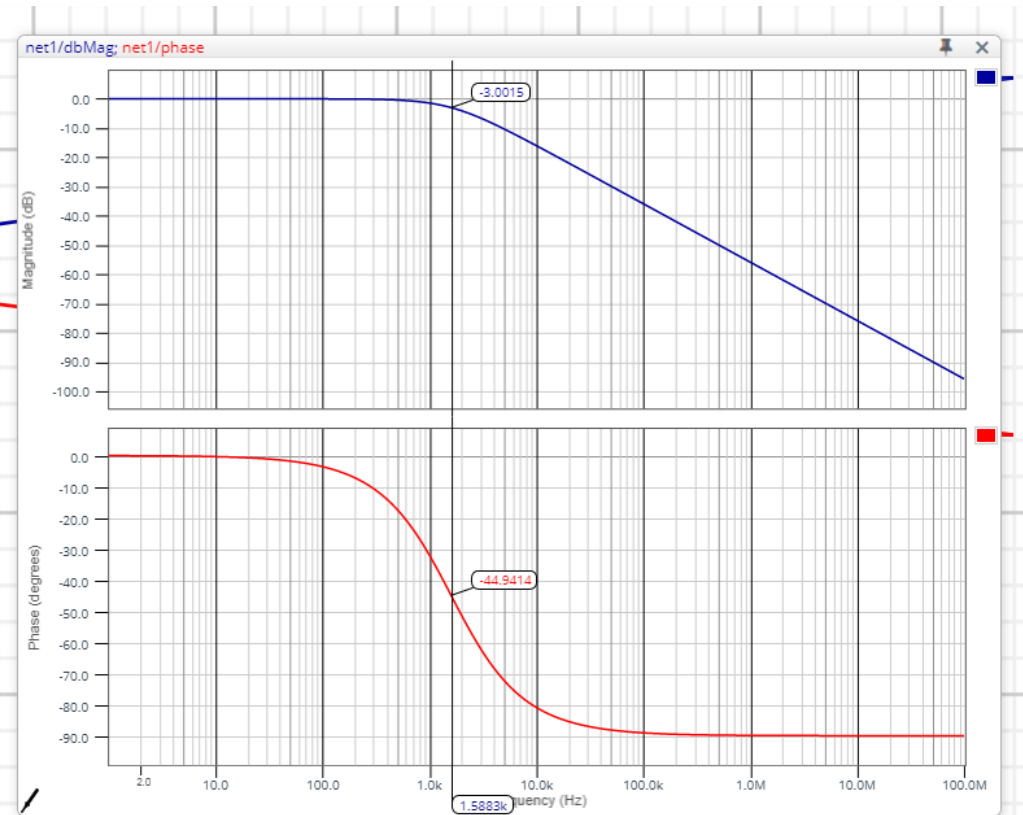
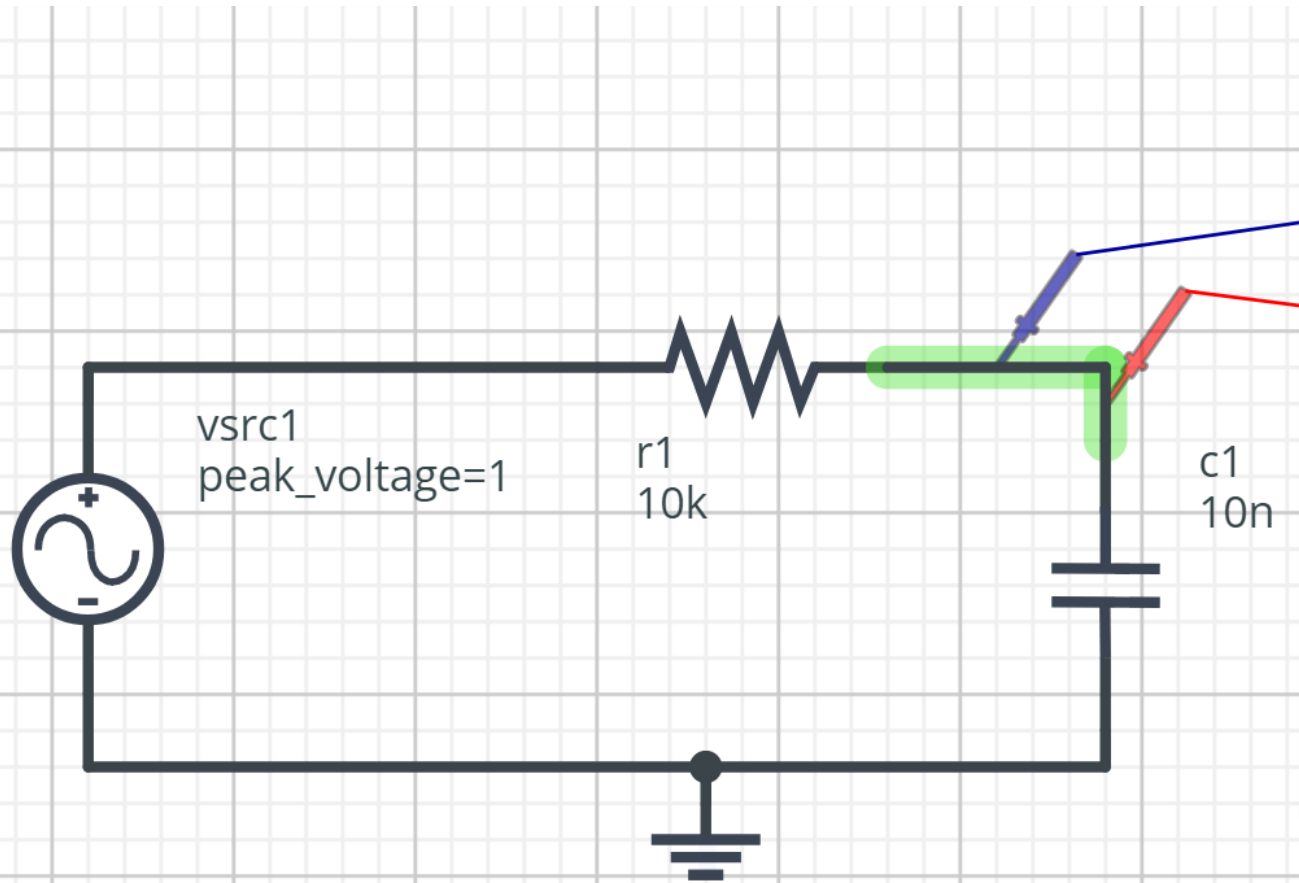
$$x \rightarrow \infty \quad \longrightarrow \quad |H(jx)| \rightarrow 0$$

$$|H(jx)|_{dB} \rightarrow -\infty$$

$$\varphi = -\frac{\pi}{2}$$



Filtrul Trece Jos

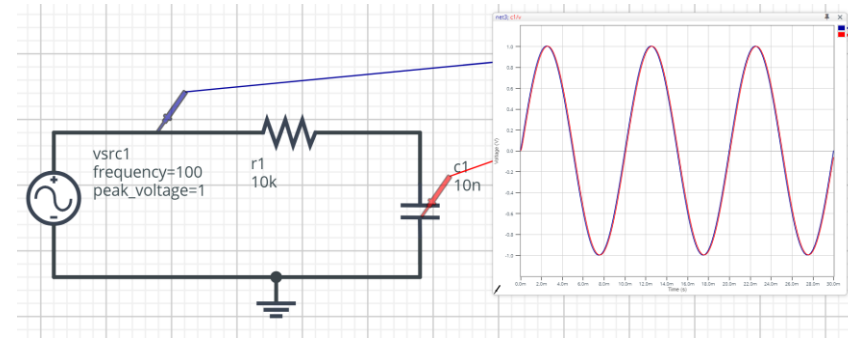


Filtrul Trece Jos

$$f_t = \frac{1}{2\pi RC} = \frac{1}{2\pi \cdot 10k\Omega \cdot 10nF} = 1591Hz$$

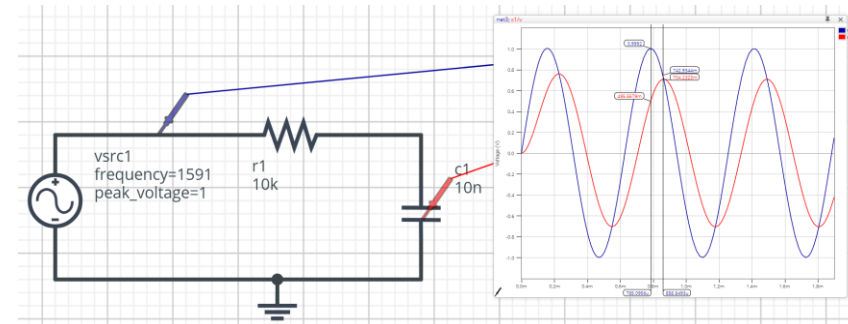
Ex1:

- semnal de intrare sinusoidal cu amplitudinea de 1V și frecvența 100Hz



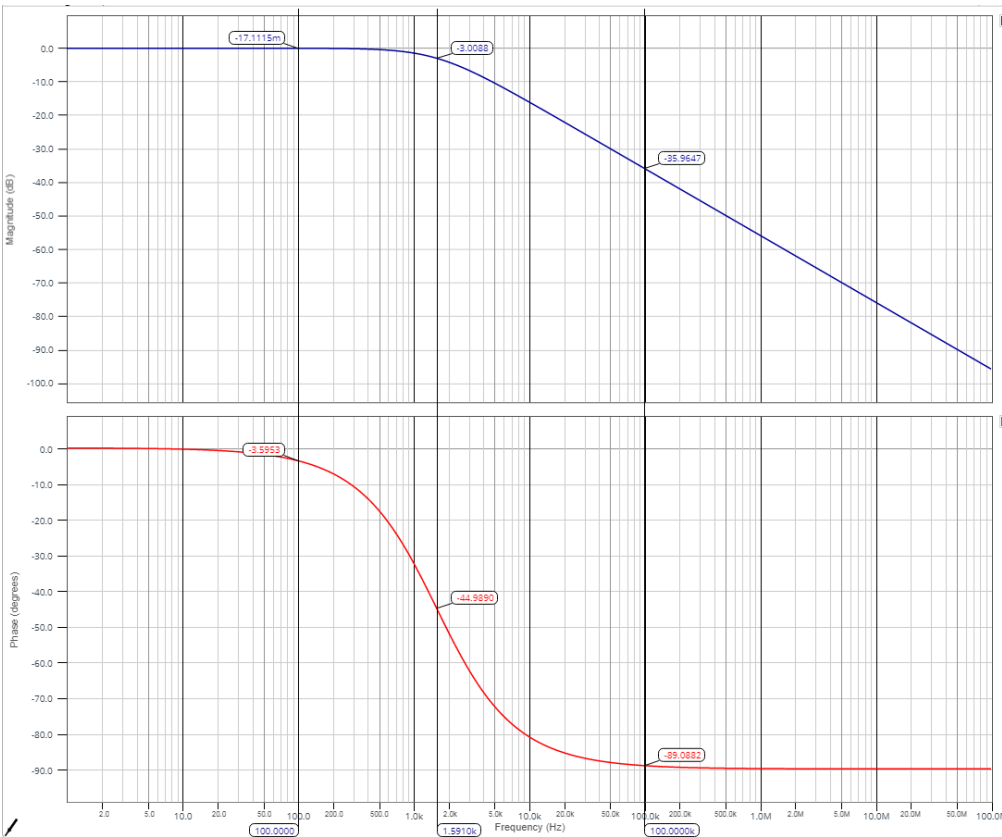
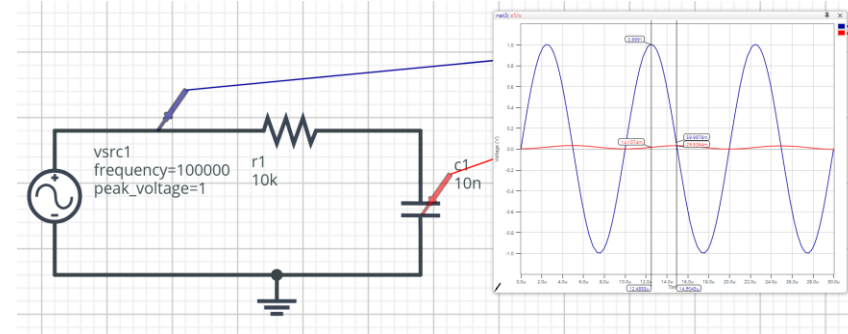
Ex2:

- semnal de intrare sinusoidal cu amplitudinea de 1V și frecvența 1591Hz



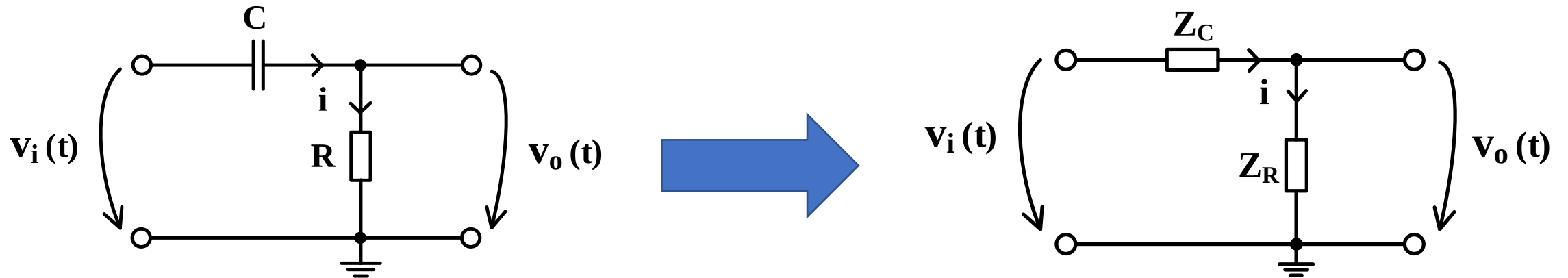
Ex3:

- semnal de intrare sinusoidal cu amplitudinea de 1V și frecvența 100kHz



Filtrul Trece Sus

I. Trecerea la impedanțe echivalente

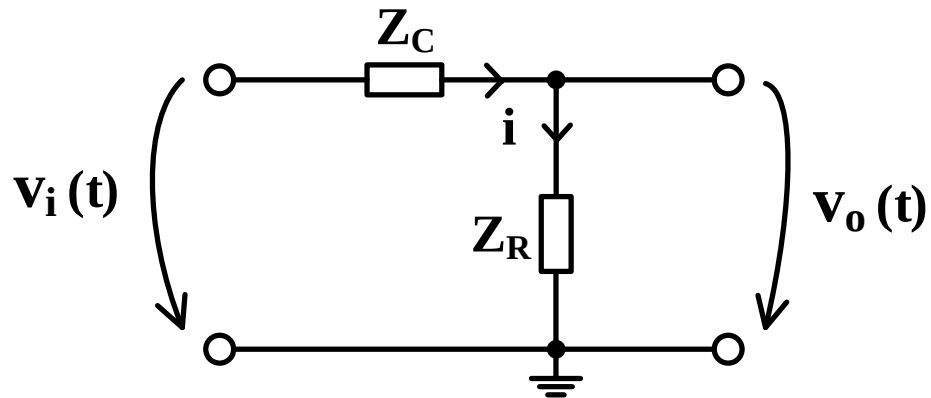


$$Z_R = R$$

$$Z_C = \frac{1}{j\omega C}$$

Filtrul Trece Sus

II. Determinarea funcției de transfer



$$\underline{H}(j\omega) = \frac{\underline{v}_o}{\underline{v}_i}$$

$$\underline{v}_o = \frac{\underline{Z}_R}{\underline{Z}_R + \underline{Z}_C} \cdot \underline{v}_i \Rightarrow \underline{H} = \frac{\underline{v}_o}{\underline{v}_i} = \frac{\underline{Z}_R}{\underline{Z}_R + \underline{Z}_C}$$

Filtrul Trece Sus

III. Înlocuirea impedanțelor

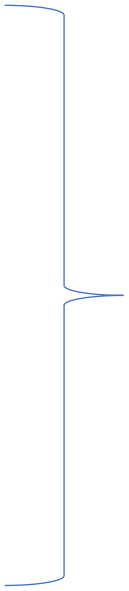
$$\Rightarrow \underline{H}(j\omega) = \frac{\underline{Z}_R}{\underline{Z}_R + \underline{Z}_C} = \frac{R}{\frac{1}{j\omega C} + R} = \frac{R}{\frac{1 + j\omega RC}{j\omega C}} = \frac{j\omega RC}{1 + j\omega RC}$$

Filtrul Trece Sus

IV. Normarea funcției de transfer

$$f_t = \frac{1}{2\pi RC}$$

$$x = \frac{f}{f_t}$$


$$\underline{H}(jx) = \frac{jx}{1 + jx}$$

Filtrul Trece Sus

V. Determinarea modulului și a fazei funcției de transfer normalată

$$\left| \underline{H(jx)} \right| = \left| \frac{jx}{1+jx} \right| = \left| \frac{(1-jx) \cdot jx}{(1+jx) \cdot (1-jx)} \right| = \left| \frac{jx + x^2}{1+x^2} \right| = \left| \frac{x^2 + jx}{1+x^2} \right| \Rightarrow \begin{cases} \left| \underline{H(jx)} \right| = \frac{\sqrt{x^4 + x^2}}{1+x^2} = \frac{x}{\sqrt{1+x^2}} \\ \varphi = \operatorname{arctg} \frac{\operatorname{Im}}{\operatorname{Re}} = \operatorname{arctg} \frac{1}{x} \end{cases}$$

Filtrul Trece Sus

VI. Reprezentarea grafică a modulului și a fazei

$$|\underline{H}|_{dB} = 20 \lg |\underline{H}|$$

$$x \rightarrow 0 \quad \longrightarrow \quad |H(jx)| \rightarrow 0$$

$$\left\{ \begin{array}{l} |H(jx)|_{dB} \rightarrow -\infty \\ \varphi = \frac{\pi}{2} \end{array} \right.$$

$$x = 1 \quad \longrightarrow \quad |H(jx)| = \frac{1}{\sqrt{2}}$$

$$\left\{ \begin{array}{l} |H(jx)|_{dB} = -3dB \\ \varphi = \frac{\pi}{4} \end{array} \right.$$

$$x \rightarrow \infty \quad \longrightarrow \quad |H(jx)| \rightarrow 1$$

$$\left\{ \begin{array}{l} |H(jx)|_{dB} \rightarrow 0 \\ \varphi = 0 \end{array} \right.$$

$$\left\{ \begin{array}{l} |H(jx)| = \frac{\sqrt{x^4 + x^2}}{1 + x^2} = \frac{x}{\sqrt{1 + x^2}} \\ \varphi = \arctg \frac{\text{Im}}{\text{Re}} = \arctg \frac{1}{x} \end{array} \right.$$

Filtrul Trece Sus

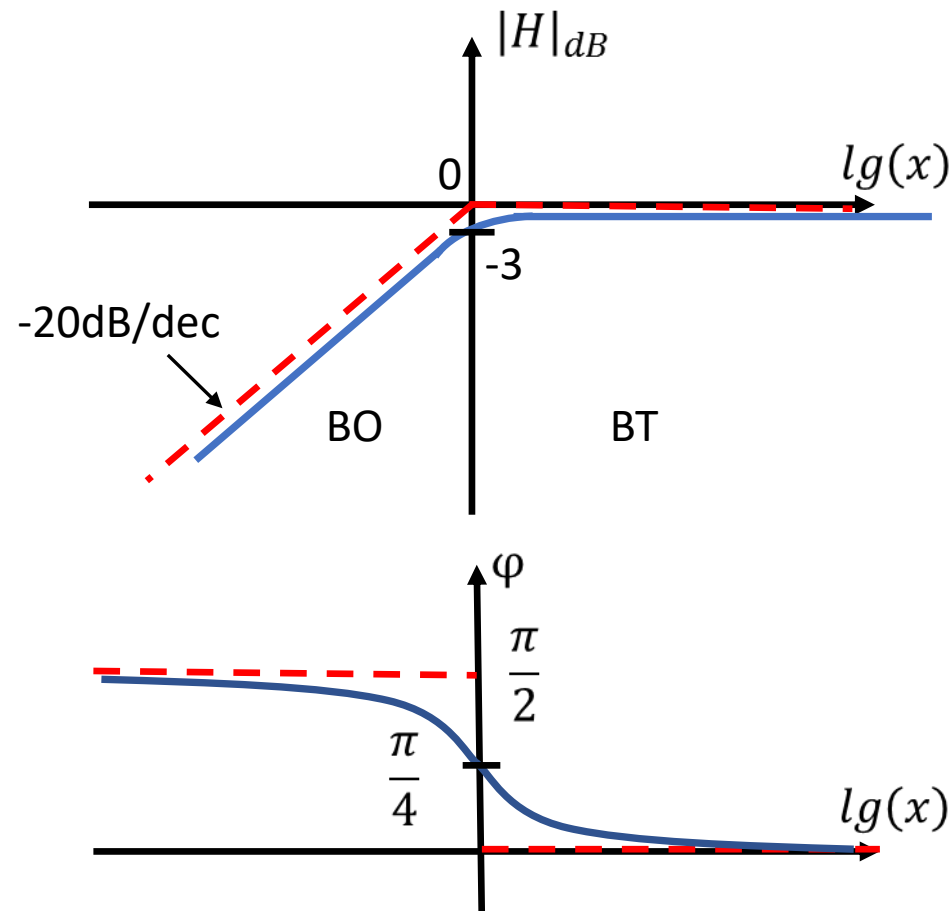
V. Determinarea modului și a fazei funcției de transfer normalată

$$\left| \underline{H}(jx) \right| = \left| \frac{jx}{1+jx} \right| = \left| \frac{(1-jx) \cdot jx}{(1+jx) \cdot (1-jx)} \right| = \left| \frac{jx + x^2}{1+x^2} \right| = \left| \frac{x^2 + jx}{1+x^2} \right| \Rightarrow \begin{cases} \left| \underline{H}(jx) \right| = \frac{\sqrt{x^4 + x^2}}{1+x^2} = \frac{x}{\sqrt{1+x^2}} \\ \varphi = \arctg \frac{\text{Im}}{\text{Re}} = \arctg \frac{1}{x} \end{cases}$$

VI. Reprezentarea grafică a modului și a fazei

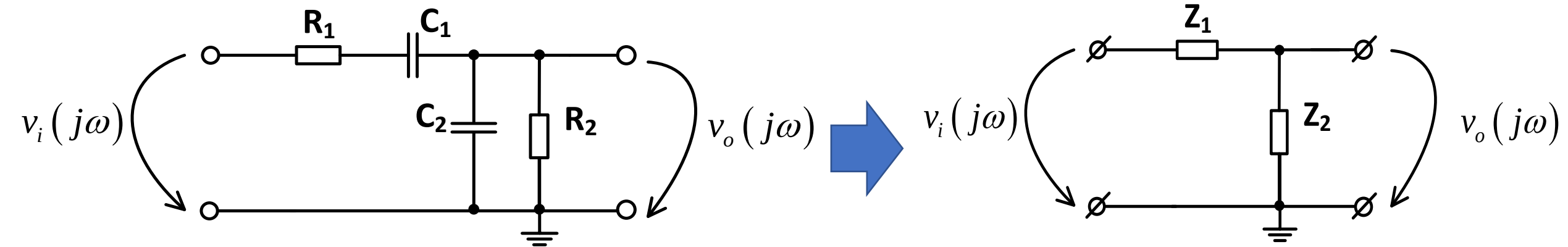
$$\left| \underline{H} \right|_{dB} = 20 \lg \left| \underline{H} \right|$$

$x \rightarrow 0$	$\longrightarrow$	$\left \underline{H}(jx) \right \rightarrow 0$	$\longrightarrow$	$\begin{cases} \left \underline{H}(jx) \right _{dB} \rightarrow -\infty \\ \varphi = \frac{\pi}{2} \end{cases}$
$x = 1$	$\longrightarrow$	$\left \underline{H}(jx) \right = \frac{1}{\sqrt{2}}$	$\longrightarrow$	$\begin{cases} \left \underline{H}(jx) \right _{dB} = -3dB \\ \varphi = \frac{\pi}{4} \end{cases}$
$x \rightarrow \infty$	$\longrightarrow$	$\left \underline{H}(jx) \right \rightarrow 1$	$\longrightarrow$	$\begin{cases} \left \underline{H}(jx) \right _{dB} \rightarrow 0 \\ \varphi = 0 \end{cases}$



Filtrul Trece Bandă (Wien)

I. Trecerea la impedanțe echivalente

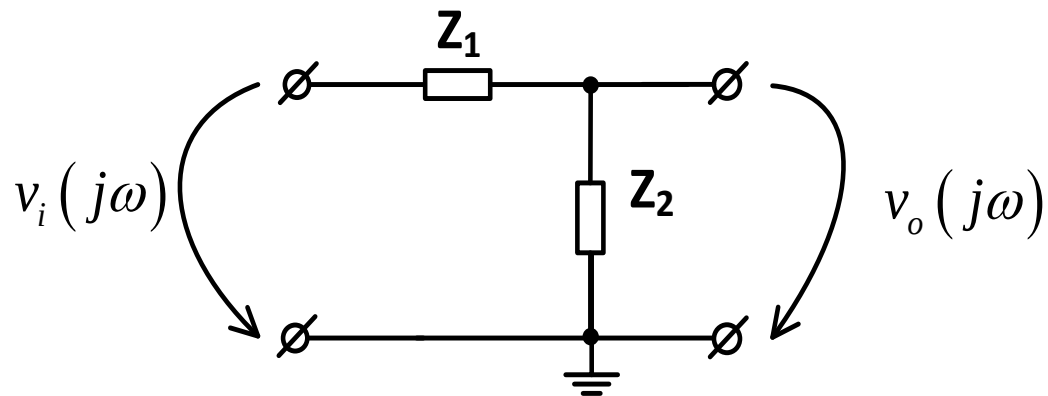


$$Z_1 = R_1 + \frac{1}{j\omega C_1} = \frac{1 + j\omega R_1 C_1}{j\omega C_1} \quad Z_2 = R_2 \parallel C_2 = \frac{R_2 \cdot \frac{1}{j\omega C_2}}{R_2 + \frac{1}{j\omega C_2}} = \frac{\frac{R_2}{j\omega C_2}}{\frac{1 + j\omega R_2 C_2}{j\omega C_2}} = \frac{R}{1 + j\omega R_2 C_2}$$

Filtrul Trece Bandă (Wien)

II. Determinarea funcției de transfer

$$\underline{H}(j\omega) = \frac{\underline{v}_o}{\underline{v}_i}$$



$$\underline{v}_o = \frac{\underline{Z}_2}{\underline{Z}_1 + \underline{Z}_2} \cdot \underline{v}_i \Rightarrow \underline{H} = \frac{\underline{v}_o}{\underline{v}_i} = \frac{\underline{Z}_2}{\underline{Z}_1 + \underline{Z}_2}$$

Filtrul Trece Bandă (Wien)

III. Înlocuirea impedanțelor

$$\begin{aligned}\Rightarrow \underline{H}(j\omega) &= \frac{\underline{Z}_2}{\underline{Z}_1 + \underline{Z}_2} = \frac{\frac{R}{1 + j\omega RC}}{\frac{1 + j\omega RC}{j\omega C} + \frac{R}{1 + j\omega RC}} = \frac{\frac{R}{1 + j\omega RC}}{\frac{(1 + j\omega RC)^2 + j\omega RC}{j\omega C \cdot (1 + j\omega RC)}} = \\ &= \frac{j\omega RC}{(1 + j\omega RC)^2 + j\omega RC} = \frac{j\omega RC}{1 - \omega^2 R^2 C^2 + 3j\omega RC}\end{aligned}$$

Filtrul Trece Bandă (Wien)

IV. Normarea funcției de transfer

$$\left. \begin{array}{l} f_0 = \frac{1}{2\pi RC} \\ x = \frac{f}{f_0} \end{array} \right\} \quad \underline{H(jx)} = \frac{jx}{1 - x^2 + 3jx} = \frac{jx \cdot (1 - x^2 - 3jx)}{(1 - x^2)^2 + 9x^2} = \frac{3x^2 + jx(1 - x^2)}{(1 - x^2)^2 + 9x^2}$$

Filtrul Trece Bandă (Wien)

V.a Determinarea modului funcției de transfer normalată

$$\begin{aligned} \left| \underline{H}(jx) \right| &= \left| \frac{3x^2 + jx(1-x^2)}{(1-x^2)^2 + 9x^2} \right| = \sqrt{\frac{9x^4 + x^2(1-x^2)^2}{\left[(1-x^2)^2 + 9x^2 \right]^2}} = \\ &= \sqrt{\frac{x^2 \left[9x^2 + (1-x^2)^2 \right]}{\left[(1-x^2)^2 + 9x^2 \right]^2}} = \frac{x}{\sqrt{9x^2 + (1-x^2)^2}} \end{aligned}$$

Filtrul Trece Bandă (Wien)

V.b Determinarea fazei funcției de transfer normalată

$$\varphi = \operatorname{arctg} \frac{\operatorname{Im}}{\operatorname{Re}} = \operatorname{arctg} \frac{1 - x^2}{3x}$$

Filtrul Trece Bandă (Wien)

VI. Reprezentarea grafică a modulului și a fazei

$$\left| H(jx) \right| = \frac{x}{\sqrt{9x^2 + (1 - x^2)^2}}$$

$$\varphi = \operatorname{arctg} \frac{1 - x^2}{3x}$$

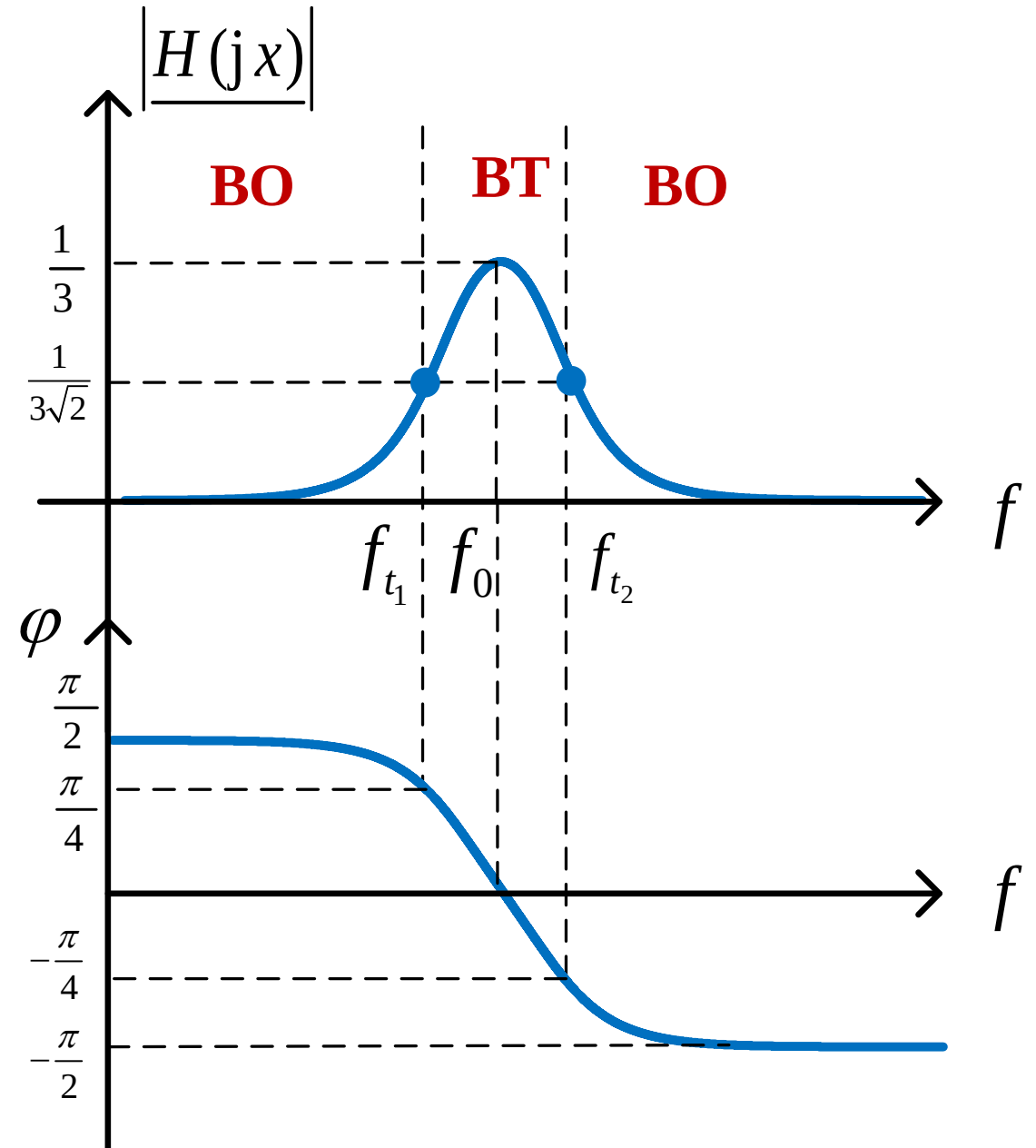
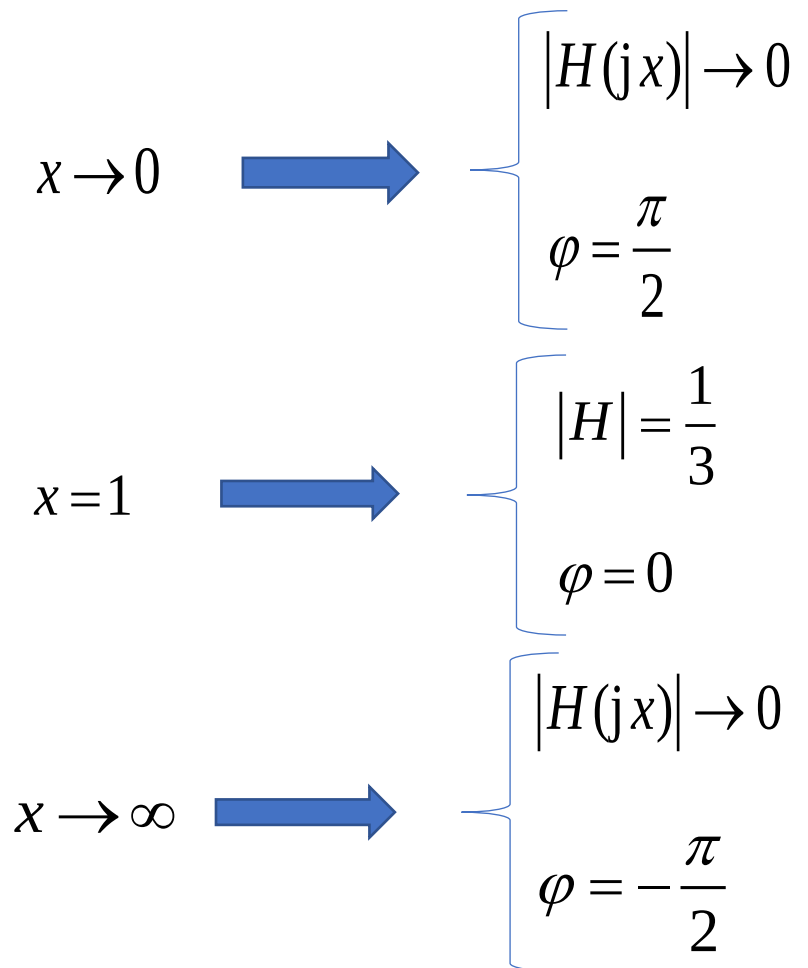
$$x \rightarrow 0 \quad \longrightarrow \quad \begin{cases} |H(jx)| \rightarrow 0 \\ \varphi = \frac{\pi}{2} \end{cases}$$

$$x = 1 \quad \longrightarrow \quad \begin{cases} |H| = \frac{1}{3} \\ \varphi = 0 \end{cases}$$

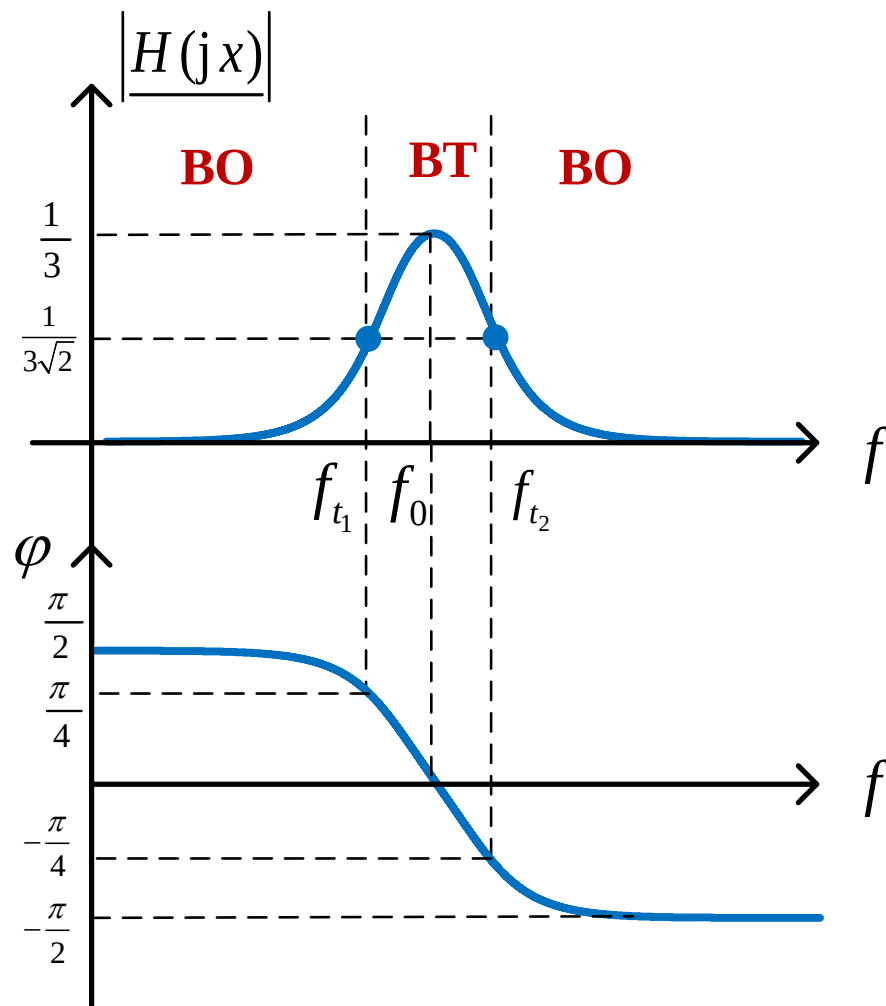
$$x \rightarrow \infty \quad \longrightarrow \quad \begin{cases} |H(jx)| \rightarrow 0 \\ \varphi = -\frac{\pi}{2} \end{cases}$$

Filtrul Trece Bandă

VI. Reprezentarea grafică a modului și a fazei



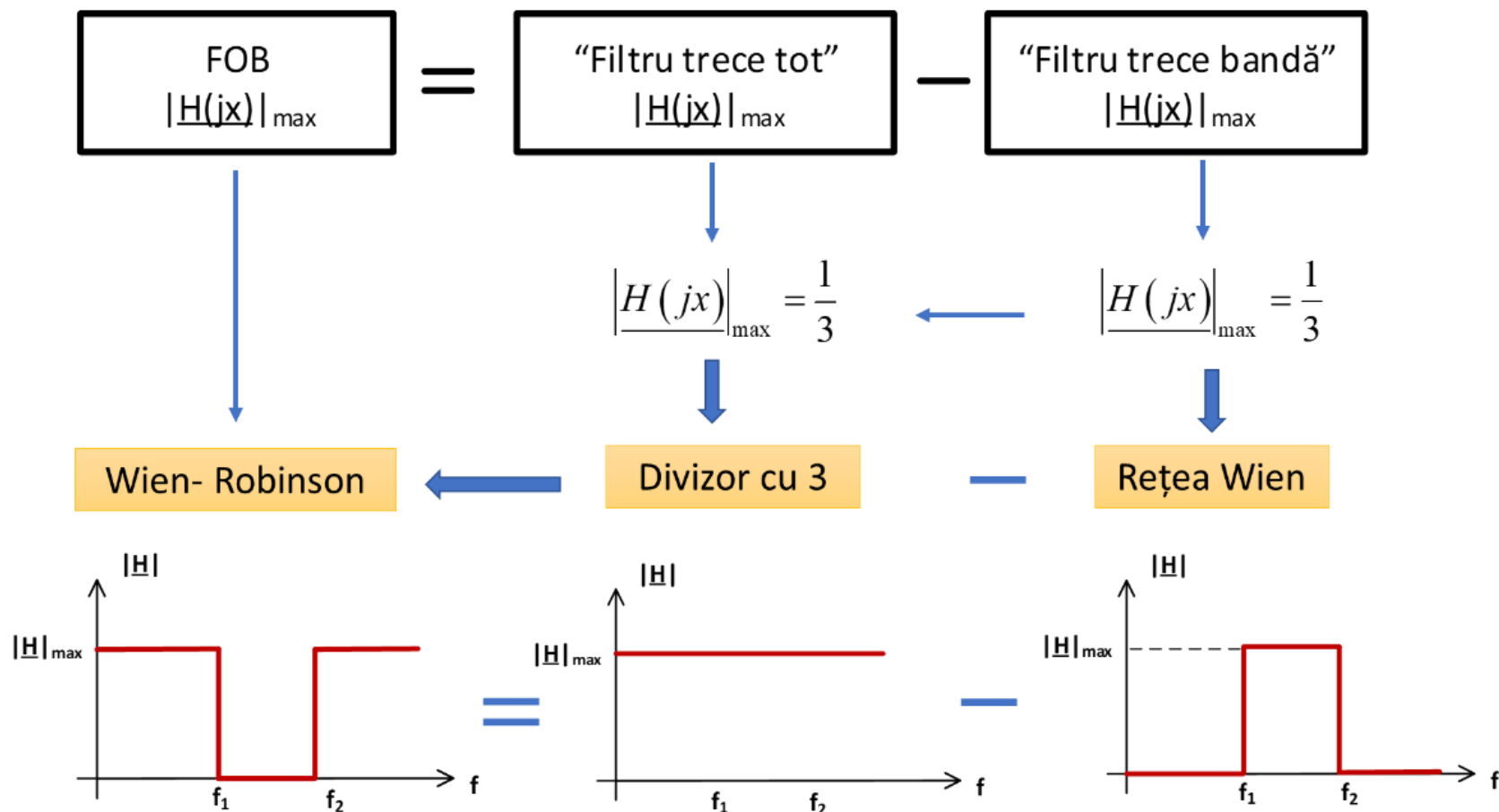
Filtrul Trece Bandă (Wien)



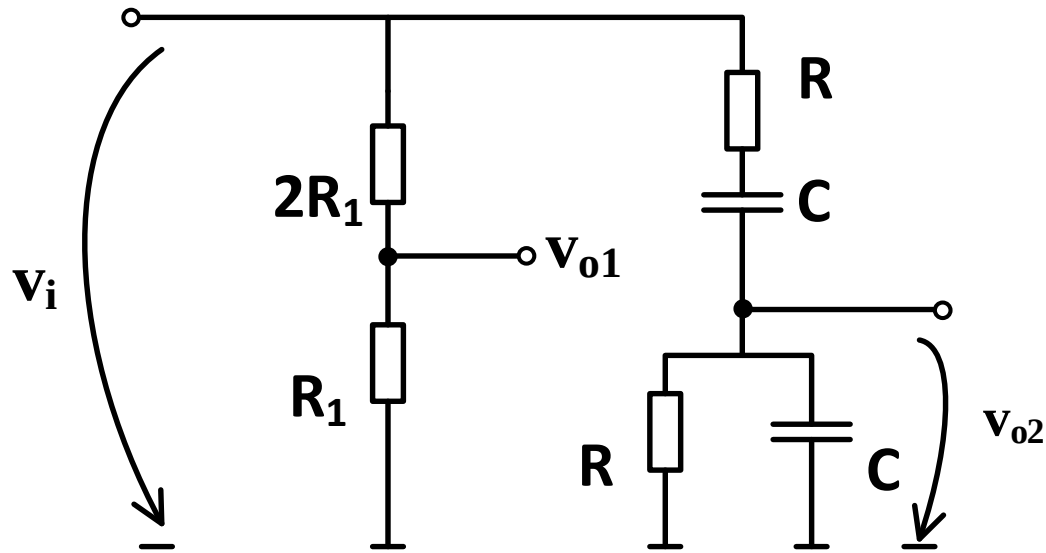
Factorul de calitate:

$$Q = \frac{f_0}{B_{3dB}}$$

Filtrul Oprește Bandă (Wien-Robinson)



Filtrul Oprește Bandă (Wien-Robinson)

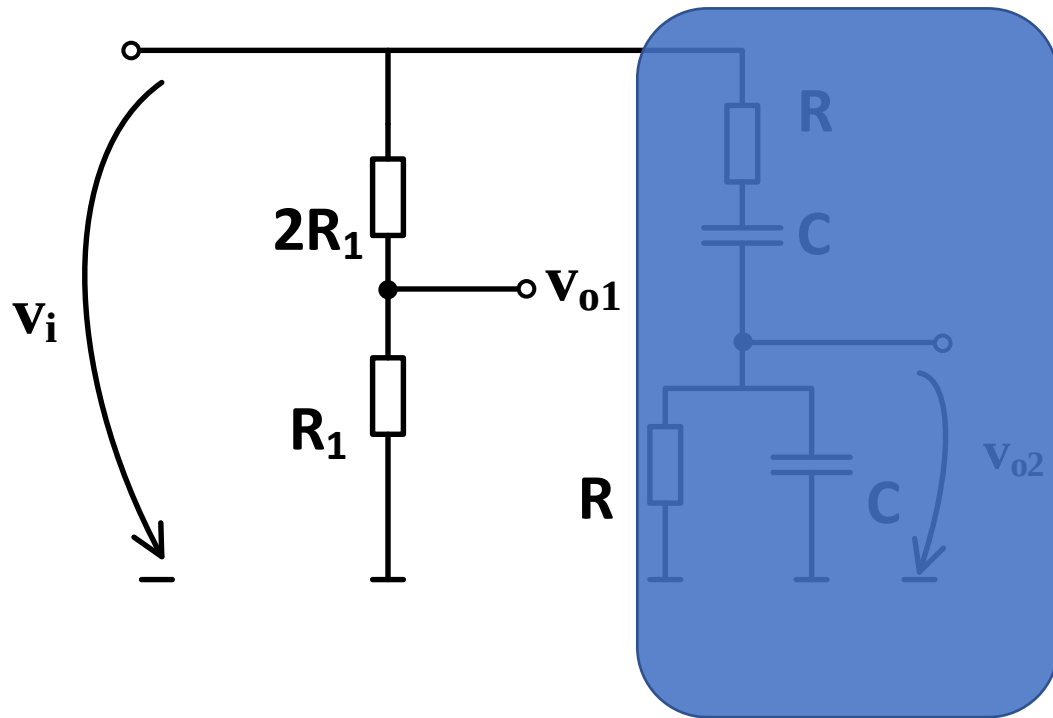


Funcția de transfer:

$$\underline{H(j\omega)} = \frac{V_o}{V_i} = \frac{V_{o1} - V_{o2}}{V_i} = \frac{V_{o1}}{V_i} - \frac{V_{o2}}{V_i}$$

Filtrul Oprește Bandă (Wien-Robinson)

Funcția de transfer:



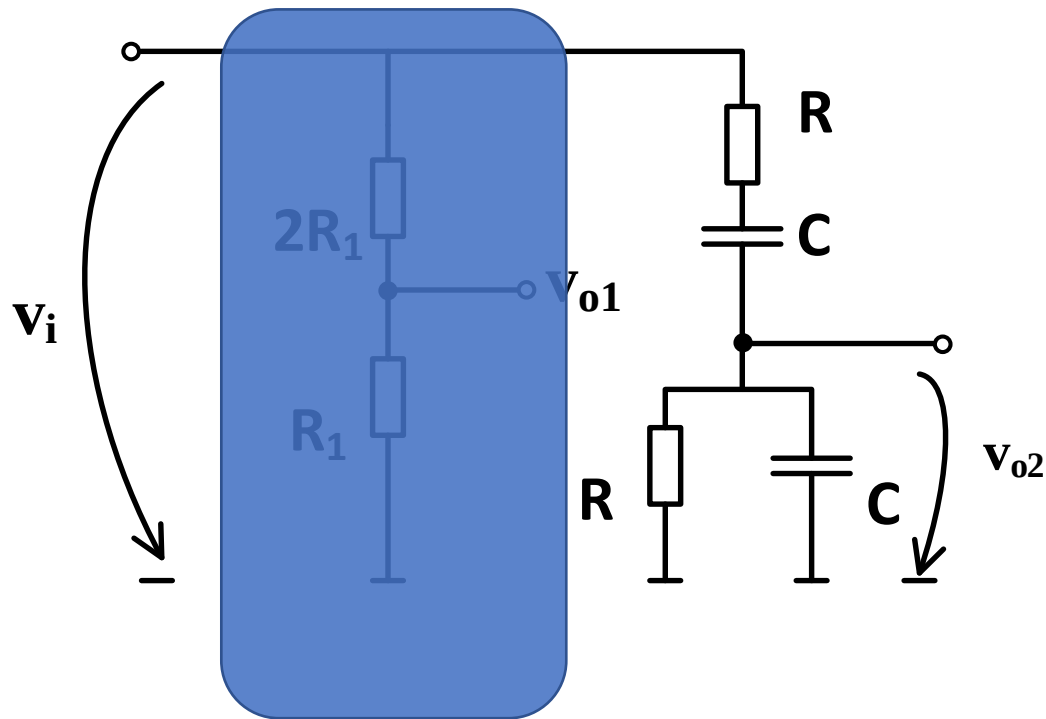
Divizorul de tensiune:

$$v_{o1} = \frac{R_1}{R_1 + 2R_1} \cdot v_i = \frac{1}{3} \cdot v_i$$

$$\boxed{\frac{v_{o1}}{v_i} = \frac{1}{3}}$$

Filtrul Oprește Bandă (Wien-Robinson)

Funcția de transfer:



FTB:

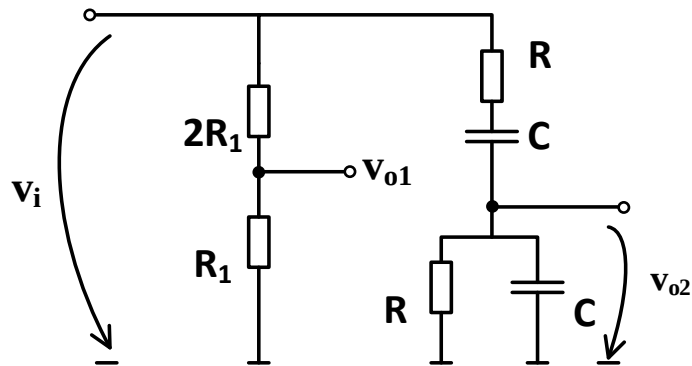
$$V_{o2} = \frac{3x^2 + jx(1 - x^2)}{(1 - x^2)^2 + 9x^2} \cdot V_i$$

$$\frac{V_{o2}}{V_i} = \frac{3x^2 + jx(1 - x^2)}{(1 - x^2)^2 + 9x^2}$$

Filtrul Oprește Bandă (Wien-Robinson)

Funcția de transfer:

$$\underline{H(j\omega)} = \frac{V_{o1}}{V_i} - \frac{V_{o2}}{V_i} = \frac{1}{3} - \frac{3x^2 + jx(1-x^2)}{(1-x^2)^2 + 9x^2} = \frac{(1-x^2)^2 + 9x^2 - 3 \cdot [3x^2 + jx(1-x^2)]}{3 \cdot [(1-x^2)^2 + 9x^2]} \Leftrightarrow$$



$$\Leftrightarrow \underline{H(j\omega)} = \frac{(1-x^2)^2 + 9x^2 - 9x^2 - 3jx(1-x^2)}{3 \cdot [(1-x^2)^2 + 9x^2]} = \frac{(1-x^2)^2 - 3jx(1-x^2)}{3 \cdot [(1-x^2)^2 + 9x^2]} \Leftrightarrow$$

$$\Leftrightarrow \underline{H(j\omega)} = \frac{(1-x^2) \cdot (1-x^2 - 3jx)}{3 \cdot [(1-x^2)^2 + 9x^2]}$$

Filtrul Oprește Bandă (Wien-Robinson)

Modulul și faza funcției de transfer:

$$\left| \underline{H(jx)} \right| = \left| \frac{(1-x^2) \cdot (1-x^2-3jx)}{3 \cdot [(1-x^2)^2 + 9x^2]} \right| = \sqrt{\frac{(1-x^2)^2 \cdot [(1-x^2)^2 + 9x^2]}{9[(1-x^2)^2 + 9x^2]^2}}$$

$$\left| \underline{H(jx)} \right| = \frac{|(1-x^2)|}{3\sqrt{(1-x^2)^2 + 9x^2}}$$

$$\varphi = \operatorname{arctg} \frac{\operatorname{Im}}{\operatorname{Re}} = \operatorname{arctg} \left(-\frac{3x}{1-x^2} \right)$$

Filtrul Oprește Bandă (Wien-Robinson)

VI. Reprezentarea grafică a modului și a fazei

$$x \rightarrow 0 \quad \left\{ \begin{array}{l} |H(jx)| = \frac{1}{3} \\ \varphi = 0 \end{array} \right.$$

$$x \rightarrow +\infty \quad \left\{ \begin{array}{l} |H(jx)| = \frac{1}{3} \\ \varphi = 0 \end{array} \right.$$

Filtrul Oprește Bandă (Wien-Robinson)

VI. Reprezentarea grafică a modului și a fazei

$$x \rightarrow 0 \quad \left\{ \begin{array}{l} |H(jx)| = \frac{1}{3} \\ \varphi = 0 \end{array} \right.$$

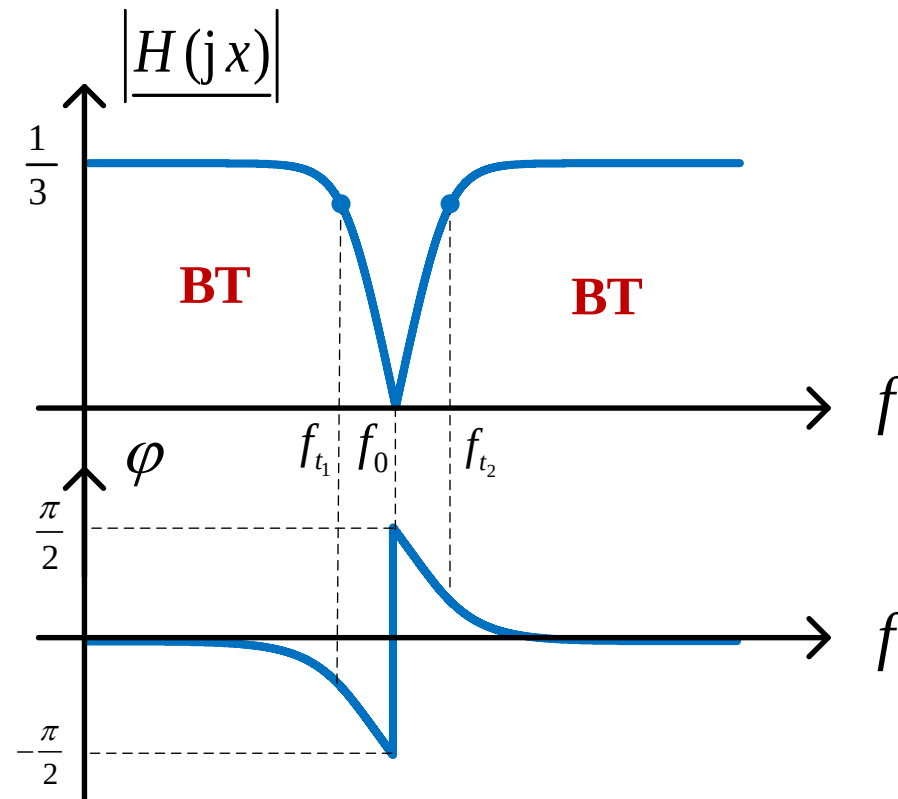
$$x \uparrow 1 \quad \left\{ \begin{array}{l} |H(jx)| = 0 \\ \varphi = -\frac{\pi}{2} \end{array} \right.$$

$$x \rightarrow +\infty \quad \left\{ \begin{array}{l} |H(jx)| = \frac{1}{3} \\ \varphi = 0 \end{array} \right.$$

$$x \downarrow 1 \quad \left\{ \begin{array}{l} |H(jx)| = 0 \\ \varphi = \frac{\pi}{2} \end{array} \right.$$

Filtrul Oprește Bandă (Wien-Robinson)

VI. Reprezentarea grafică a modului și a fazei



Factorul de calitate:

$$Q = \frac{f_0}{B_{3dB}}$$